USE CASE SCENARIOS

Project: Bank Management System

Use Case 1: Authenticate User

Primary Actor: Customer, Bank Employee, Loan Officer, Bank Manager, System Admin, Auditor

Purpose: Let any registered role log in securely so the system can enforce role-based access before granting further actions.

Preconditions

- The user has a valid registered account and credentials.
- The system's authentication service is reachable.

Postconditions

- On success, the user has an active session scoped to their role.
- On failure, no session is created and the login attempt is logged.

Main Success Scenario

1. User enters their username and password.
2. System validates the credentials against stored records.
3. System checks the user's role and assigns the matching permissions.
4. System creates a session and redirects the user to their dashboard.
5. Use case ends successfully.

Alternate Flow — Invalid Credentials

2a. System cannot match the entered credentials to a valid account.

a) System displays an error message without indicating whether the username or password was wrong.
b) User re-enters credentials.
c) After 5 failed attempts, the account is temporarily locked and the user is notified.

Relationship Used in the UML Model

Authenticate User *«includes»* Enforce RBAC (NFR-001), since every login must resolve the user's role before any further use case is allowed.

Use Case 2: Deposit / Withdraw Funds

Primary Actor: Customer, Bank Employee

Purpose: Move money into or out of a customer's account with the balance updating immediately.

Preconditions

- The user is authenticated and authorized for this account.
- The account is active and not frozen.

Postconditions

- On success, the balance is updated and a confirmation is shown.
- On failure, the balance is unchanged and the user is told why.

Main Success Scenario

1. User selects deposit or withdrawal and enters an amount.

2. System validates the amount against account rules (e.g. sufficient funds for withdrawal).
3. System updates the balance and records the transaction.
4. System shows a confirmation with the new balance.
5. Use case ends successfully.

Alternate Flow — Insufficient Funds

2a. Withdrawal amount exceeds the available balance.

a) System rejects the withdrawal and shows the current balance.
b) User enters a lower amount or cancels.

Relationship Used in the UML Model

Deposit / Withdraw Funds *«extends»* is the base for Monitor Performance (NFR-002), which extends it to track response time against the 2–3 second target whenever a transaction runs.

Use Case 3: Transfer Money

Primary Actor: Customer

Purpose: Move funds from the customer's account to another account within the bank.

Preconditions

- The user is authenticated.
- The sender account has sufficient balance.
- The receiver account exists and is active.

Postconditions

- On success, both sender and receiver balances update correctly.
- On failure, no balance changes on either side.

Main Success Scenario

1. Customer enters the receiver's account number and the amount.
2. System verifies the receiver account exists and the sender has sufficient funds.
3. System debits the sender and credits the receiver in a single transaction.
4. System shows a confirmation to the customer.
5. Use case ends successfully.

Alternate Flow — Transfer Fails Midway

3a. System detects a failure after debiting the sender but before crediting the receiver.

a) System automatically rolls back the debit.
b) System shows an error and logs the failed attempt for review.
c) Customer is asked to retry the transfer.

Relationship Used in the UML Model

Transfer Money *«includes»* Authenticate User, since a transfer can only be initiated from an active, verified session.

Use Case 4: Apply for Loan

Primary Actor: Customer (applies), Loan Officer (reviews)

Purpose: Let a customer submit a loan request and a loan officer review, approve, or reject it.

Preconditions

- The customer is authenticated and has an active account.
- The loan officer has access to the review queue.

Postconditions

- On success, the application is submitted with a reference ID and appears in the officer's queue.
- On failure, the application is not created and the customer is notified.

Main Success Scenario

1. Customer fills out the loan application form and submits it.
2. System validates the form and generates a reference ID.
3. System places the application in the loan officer's review queue.
4. Loan officer reviews the application and approves or rejects it.
5. System notifies the customer of the decision.
6. Use case ends successfully.

Alternate Flow — Incomplete Application

1a. Customer submits the form with required fields missing.

a) System highlights the missing fields and blocks submission.
b) Customer completes the fields and resubmits.

Relationship Used in the UML Model

Apply for Loan *«includes»* Authenticate User, and Review Loan Applications is a separate use case triggered once an application enters the queue.